

**SE – LAB 1**

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**PROBLEM STATEMENT : 13- Patient Health Record Consent Management System**

| Req ID | Type       | Description                                                                                                                                          | Priority | Acceptance Criteria                                                                                                                           | Rationale                                                         |
|--------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| FR-001 | Functional | The system shall allow patients to grant time-bounded (e.g., 24-hour) access permissions for specific diagnostic records to verified clinic doctors. | High     | <b>Pass:</b> Clinic access automatically revokes upon expiry timestamp.<br><br><b>Fail:</b> Clinic accesses records after consent expiration. | Core functionality enabling patient-centric, secure data sharing. |

|               |            |                                                                                                                                                                 |        |                                                                                                                                                                                                  |                                                                                 |
|---------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>FR-002</b> | Functional | The system shall allow a patient to manually revoke any active consent permission before the scheduled expiration time.                                         | High   | <p><b>Pass:</b> A doctor's request to access records returns an immediate 403 Forbidden error upon revocation.</p> <p><b>Fail:</b> Doctor continues to view records after manual revocation.</p> | Enforces patient autonomy and immediate control over sensitive medical records. |
| <b>FR-003</b> | Functional | The system shall allow clinic administrators to verify and approve registered medical staff credentials before granting them access to request patient records. | Medium | <p><b>Pass:</b> Unverified medical staff cannot initiate consent requests or receive access.</p> <p><b>Fail:</b> Unverified practitioner accesses patient data.</p>                              | Protects patient privacy by preventing fraudulent or unaccredited access.       |

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| <b>FR-004</b> | Functional | The system shall provide patients with an active consent dashboard displaying the real-time status (Active, Expired, Revoked) of all permission grants.                     | Medium | <p><b>Pass:</b> Dashboard updates within 1 second of any status change showing correct timestamps.</p> <p><b>Fail:</b> Revoked or expired grants remain marked as active on the dashboard.</p>                              | Provides transparency and full visibility to patients over their data lifecycle. |
| <b>FR-005</b> | Functional | The system shall enforce granular categorization filters, allowing patients to select specific record types (e.g., Blood Reports, Imaging) to share while excluding others. | High   | <p><b>Pass:</b> A doctor granted access to Lab Reports receives an access denied error when attempting to fetch Cardiology reports.</p> <p><b>Fail:</b> All record types are exposed regardless of selected categories.</p> | Prevents over-exposure of sensitive and unrelated medical history.               |

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| <b>NFR-001</b> | Non-Functional (Security & Audit) | All consent grant, revoke, and access events shall be permanently written to an append-only audit trail.                                        | High | <p><b>Pass:</b> Benchmarking tests confirm target latency and security standards under simulated peak load with zero log mutability.</p> <p><b>Fail:</b> Any consent event fails to generate an immutable log entry.</p>                | Guarantees non-repudiation and strict compliance with health data privacy regulations.              |
| <b>NFR-002</b> | Non-Functional (Performance)      | The system shall validate consent authorization tokens and return an access decision in under 200 ms under a concurrent load of 1,000 requests. | High | <p><b>Pass:</b> 95th percentile response latency is <math>\leq 200 \text{ ms}</math> under simulated load testing.</p> <p><b>Fail:</b> Response latency exceeds <math>200 \text{ ms}</math> or the authorization gateway times out.</p> | Ensures healthcare professionals can access time-critical patient data without latency bottlenecks. |

